DATA ANALYTICS / PORTFOLIO PROJECT

# E-Commerce Market Basket Analysis

Understanding product associations and turning transaction data into actionable business strategies.

TRANSACTION ANALYSIS • ASSOCIATION RULES • BUSINESS INSIGHTS

PROJECT CASE STUDY

**83K+**

transactions analyzed

# Project overview

## The business challenge

Understand which products are commonly purchased together and translate those patterns into practical opportunities for cross-selling, product recommendations, and selective bundling.

## Project objectives

- Identify top products and revenue-leading categories.
- Understand the distribution of basket sizes.
- Discover frequent product combinations and association rules.
- Develop measurable business recommendations.

**TRANSACTIONS**

**83,014**

**UNIQUE PRODUCTS**

**2,412**

**PRODUCT CATEGORIES**

**86**

**TOTAL REVENUE**

**17.56M**

Dashboard value; currency unspecified

# Analytical approach

### 01 Transaction overview

Review transaction, product, category and revenue KPIs.

### 02 Basket size analysis

Measure how many items are included in each transaction.

### 03 Frequent itemsets

Identify product combinations that appear together.

### 04 Association rules

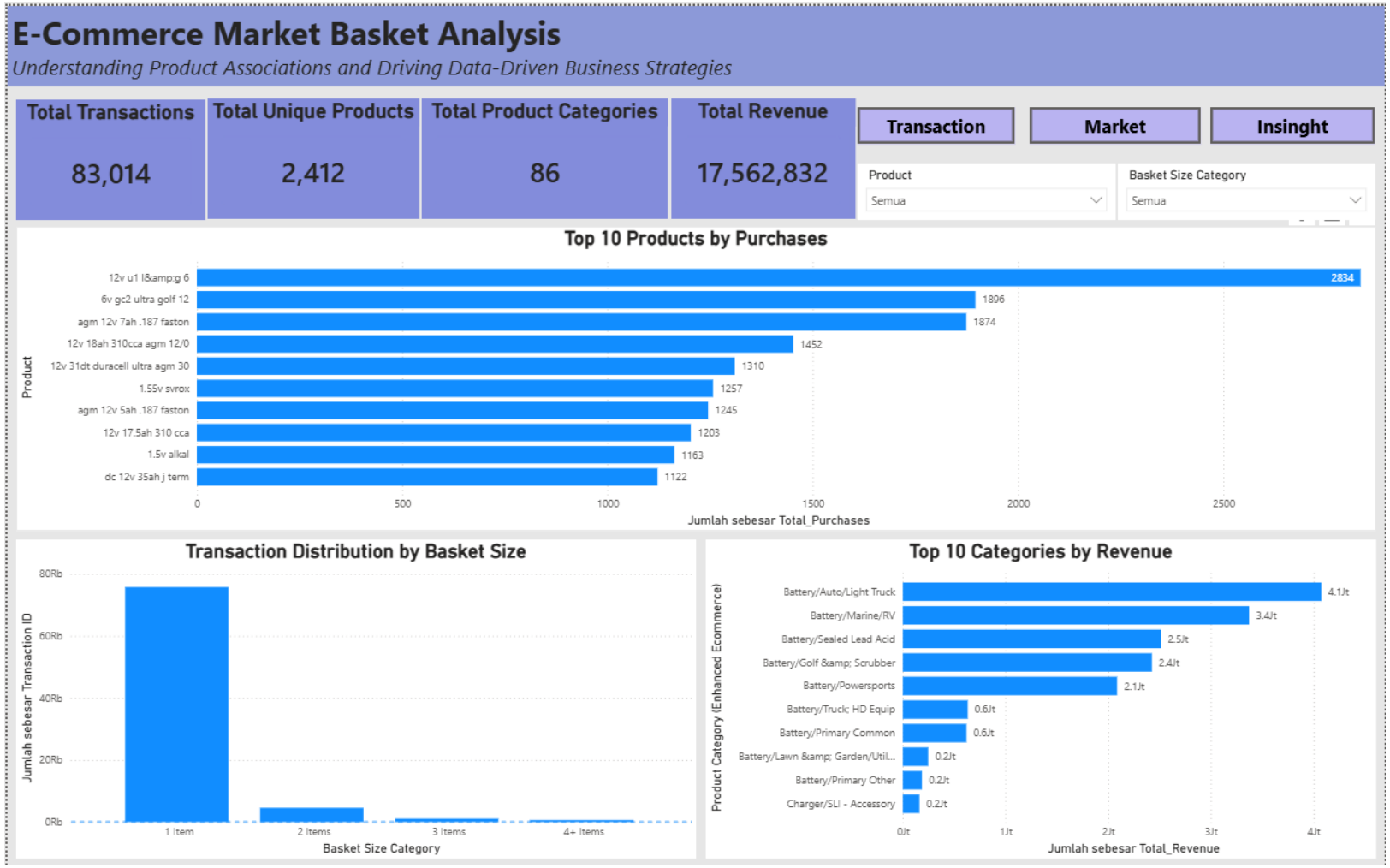
Evaluate support, confidence and lift.

### 05 Business translation

Convert patterns into insights and testable recommendations.

**Association rule metrics** Support = frequency | Confidence = conditional likelihood | Lift = strength relative to baseline purchase probability

# Transaction performance




## What stands out

- Top purchased item: 12v u1 1&8amp;g 6 with 2,834 purchases.
- Battery/Auto/Light Truck leads visible category revenue at about 4.1M.
- One-item baskets dominate the transaction distribution.

## Business implication

Test relevant add-on recommendations to encourage multi-item baskets.

# From transaction patterns to opportunities

TOP PRODUCT PURCHASES

**2,834**

12v u1 1&8amp;g 6

TOP CATEGORY REVENUE

**~4.1M**

Battery/Auto/Light Truck

DOMINANT BASKET SIZE

**1 item**

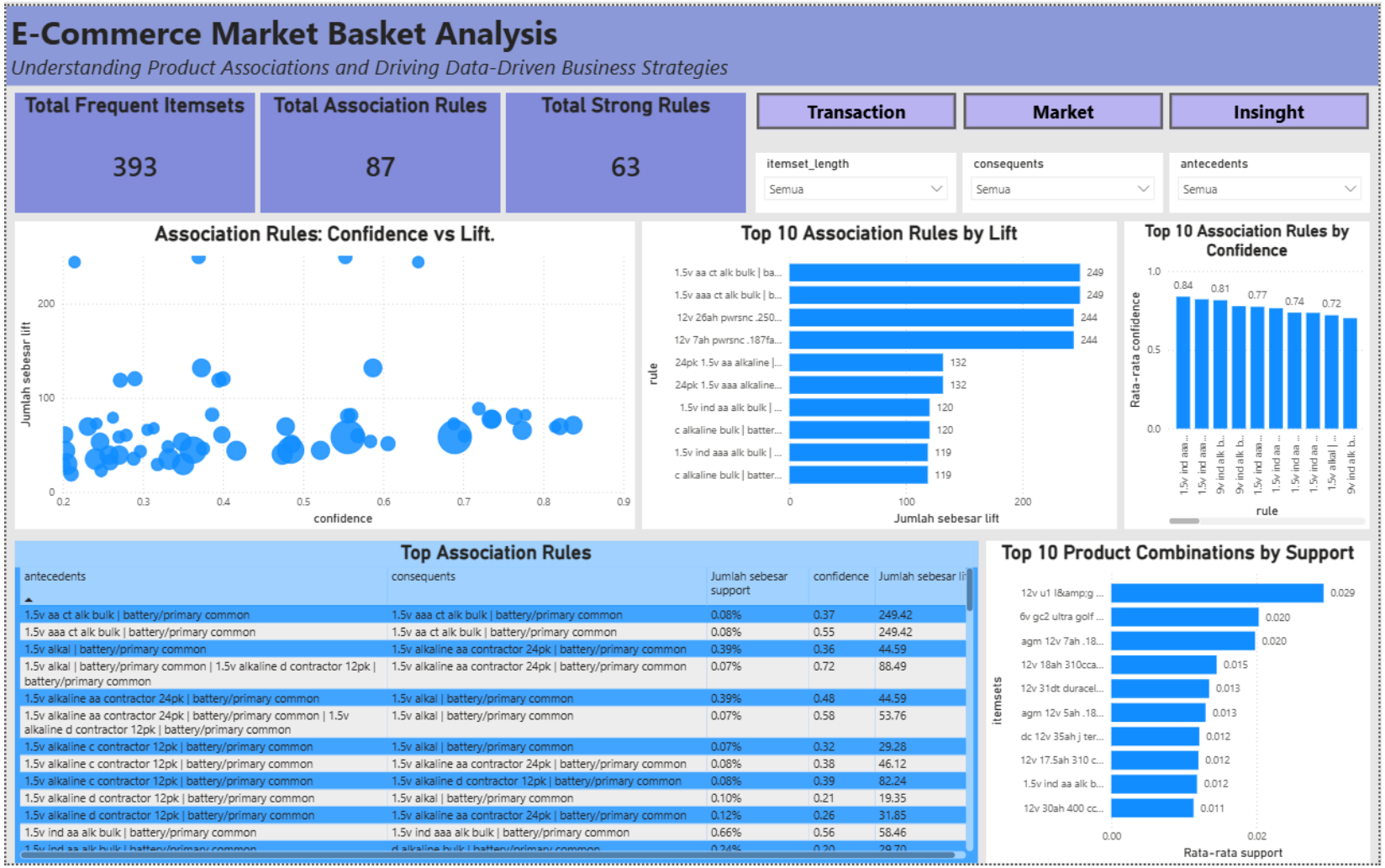
Multi-item baskets are less common

## Interpretation

- High-frequency battery products are candidates for merchandising focus.
- Revenue concentration in battery categories suggests inventory and promotion opportunities.
- The prevalence of single-item transactions creates a hypothesis: relevant complementary products may increase basket size.

**Important: this is an opportunity to test, not proof that recommendations will increase sales.**

# Product association results




**FREQUENT ITEMSETS**

**393**

**ASSOCIATION RULES**

**87**

**STRONG RULES**

**63**

### Selected displayed rule examples

Support 0.66% | Confidence 55.57% | Lift 58.46

# How to interpret association rules

## SUPPORT

**How often does this product combination appear across all transactions?**

Higher support can indicate broader reach.

## CONFIDENCE

**When customers buy the antecedent, how often do they also buy the consequent?**

Useful for estimating recommendation relevance.

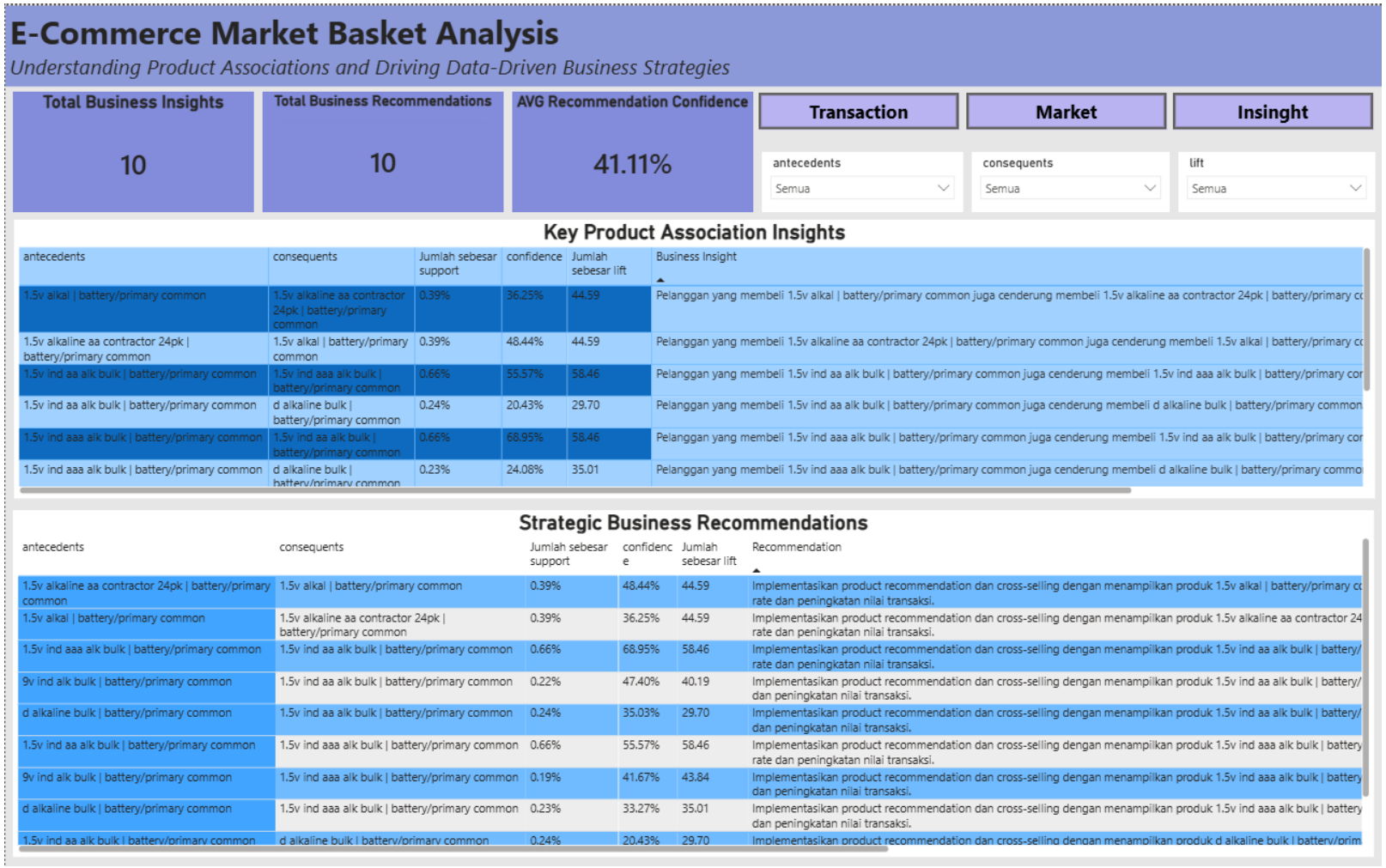
## LIFT

**How much more likely is the consequent given the antecedent, compared with its baseline likelihood?**

Lift above 1 indicates positive association.

**A high lift alone is not enough: review support, confidence, transaction count, margin, and stock availability before activation.**

# Business insights and recommendations



## E-Commerce Market Basket Analysis

*Understanding Product Associations and Driving Data-Driven Business Strategies*

| Total Business Insights | Total Business Recommendations | AVG Recommendation Confidence |
|---|---|---|
| 10 | 10 | 41.11% |

Transaction | Market | Insinght

antecedents: Semua | consequents: Semua | lift: Semua

### Key Product Association Insights

| antecedents | consequents | Jumlah sebesar support | confidence | Jumlah sebesar lift | Business Insight |
|---|---|---|---|---|---|
| 1.5v alkal \| battery/primary common | 1.5v alkaline aa contractor 24pk \| battery/primary common | 0.39% | 36.25% | 44.59 | Pelanggan yang membeli 1.5v alkal \| battery/primary common juga cenderung membeli 1.5v alkaline aa contractor 24pk \| battery/primary co |
| 1.5v alkaline aa contractor 24pk \| battery/primary common | 1.5v alkal \| battery/primary common | 0.39% | 48.44% | 44.59 | Pelanggan yang membeli 1.5v alkaline aa contractor 24pk \| battery/primary common juga cenderung membeli 1.5v alkal \| battery/primary co |
| 1.5v ind aa alk bulk \| battery/primary common | 1.5v ind aaa alk bulk \| battery/primary common | 0.66% | 55.57% | 58.46 | Pelanggan yang membeli 1.5v ind aa alk bulk \| battery/primary common juga cenderung membeli 1.5v ind aaa alk bulk \| battery/primary cor |
| 1.5v ind aa alk bulk \| battery/primary common | d alkaline bulk \| battery/primary common | 0.24% | 20.43% | 29.70 | Pelanggan yang membeli 1.5v ind aa alk bulk \| battery/primary common juga cenderung membeli d alkaline bulk \| battery/primary common |
| 1.5v ind aaa alk bulk \| battery/primary common | 1.5v ind aa alk bulk \| battery/primary common | 0.66% | 68.95% | 58.46 | Pelanggan yang membeli 1.5v ind aaa alk bulk \| battery/primary common juga cenderung membeli 1.5v ind aa alk bulk \| battery/primary cor |
| 1.5v ind aaa alk bulk \| battery/primary common | d alkaline bulk \| battery/primary common | 0.23% | 24.08% | 35.01 | Pelanggan yang membeli 1.5v ind aaa alk bulk \| battery/primary common juga cenderung membeli d alkaline bulk \| battery/primary commo |

### Strategic Business Recommendations

| antecedents | consequents | Jumlah sebesar support | confidence | Jumlah sebesar lift | Recommendation |
|---|---|---|---|---|---|
| 1.5v alkaline aa contractor 24pk \| battery/primary common | 1.5v alkal \| battery/primary common | 0.39% | 48.44% | 44.59 | Implementasikan product recommendation dan cross-selling dengan menampilkan produk 1.5v alkal \| battery/primary co rate dan peningkatan nilai transaksi. |
| 1.5v alkal \| battery/primary common | 1.5v alkaline aa contractor 24pk \| battery/primary common | 0.39% | 36.25% | 44.59 | Implementasikan product recommendation dan cross-selling dengan menampilkan produk 1.5v alkaline aa contractor 24 rate dan peningkatan nilai transaksi. |
| 1.5v ind aaa alk bulk \| battery/primary common | 1.5v ind aa alk bulk \| battery/primary common | 0.66% | 68.95% | 58.46 | Implementasikan product recommendation dan cross-selling dengan menampilkan produk 1.5v ind aa alk bulk \| battery/ rate dan peningkatan nilai transaksi. |
| 9v ind alk bulk \| battery/primary common | 1.5v ind aa alk bulk \| battery/primary common | 0.22% | 47.40% | 40.19 | Implementasikan product recommendation dan cross-selling dengan menampilkan produk 1.5v ind aa alk bulk \| battery/ dan peningkatan nilai transaksi. |
| d alkaline bulk \| battery/primary common | 1.5v ind aa alk bulk \| battery/primary common | 0.24% | 35.03% | 29.70 | Implementasikan product recommendation dan cross-selling dengan menampilkan produk 1.5v ind aa alk bulk \| battery/ dan peningkatan nilai transaksi. |
| 1.5v ind aa alk bulk \| battery/primary common | 1.5v ind aaa alk bulk \| battery/primary common | 0.66% | 55.57% | 58.46 | Implementasikan product recommendation dan cross-selling dengan menampilkan produk 1.5v ind aaa alk bulk \| battery rate dan peningkatan nilai transaksi. |
| 9v ind alk bulk \| battery/primary common | 1.5v ind aaa alk bulk \| battery/primary common | 0.19% | 41.67% | 43.84 | Implementasikan product recommendation dan cross-selling dengan menampilkan produk 1.5v ind aaa alk bulk \| battery dan peningkatan nilai transaksi. |
| d alkaline bulk \| battery/primary common | 1.5v ind aaa alk bulk \| battery/primary common | 0.23% | 33.27% | 35.01 | Implementasikan product recommendation dan cross-selling dengan menampilkan produk 1.5v ind aaa alk bulk \| battery dan peningkatan nilai transaksi. |
| 1.5v ind aa alk bulk \| battery/primary common | d alkaline bulk \| battery/primary common | 0.24% | 20.43% | 29.70 | Implementasikan product recommendation dan cross-selling dengan menampilkan produk d alkaline bulk \| battery/prim |

## Insight highlights

- 10 business insights generated.
- 10 strategic recommendations listed.
- Average recommendation confidence: 41.11%.
- Battery-related product combinations appear repeatedly.

## Recommended direction

Pilot product recommendations and selective cross-selling; validate the effect through controlled experiments.

# Strategic recommendations

| | | |
|---|---|---|
| 01 | **Personalized recommendations** | Show complementary products on product and cart pages when a relevant antecedent is selected. |
| 02 | **Cross-selling pilot** | Start with rules that balance support, confidence and lift; compare with a control group. |
| 03 | **Selective bundles** | Test a limited number of battery-related product bundles and measure incremental profit. |
| 04 | **Merchandising focus** | Prioritize high-revenue categories while monitoring margin, inventory and customer demand. |
| 05 | **Continuous evaluation** | Refresh rules periodically and monitor recommendation performance. |

# Measuring business impact

## Success metrics to monitor

- ✓ **Average Basket Size** — Average number of items per transaction.
- ✓ **Attach Rate** — Share of transactions that include the recommended add-on.
- ✓ **Recommendation CTR** — Share of users who click a recommendation.
- ✓ **Conversion Rate** — Share of recommendation clicks that result in a purchase.
- ✓ **Revenue & Profit / Transaction** — Commercial impact beyond item count.
- ✓ **Incremental Revenue** — Difference between test and control groups.

## Conclusion

**The analysis reveals product associations that can inform smarter recommendations and cross-selling. The next step is to validate their commercial impact through A/B testing, with basket size, conversion, revenue and profit as the key measures.**

*Data note: period, currency, rule thresholds, and exact algorithm are not visible in the supplied dashboard screenshots; confirm these details from the source dataset and notebook before publishing.*